RAGHAV UPADHYAY

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SUMMARY

AI/ML Engineer building production-grade LLM and deep learning systems — from retrieval to evaluation to deployment. Experienced with RAG pipelines, hybrid retrieval, LLM-as-a-judge evaluation wired into CI, and uncertainty-aware deep learning. MS in Data Science, University of Arizona (May 2026). Open to full-time AI/ML Engineer roles across the US.

TECHNICAL SKILLS

- **LLMs & RAG:** OpenAI GPT-4.1, LLaMA (Groq API), Hugging Face Transformers, RAG pipelines, ChromaDB, sentence-transformers, hybrid search (vector + BM25), cross-encoder reranking, prompt engineering, citation grounding
- **LLM Evaluation & Reliability:** LLM-as-a-judge, hallucination detection, faithfulness/relevance/citation metrics, CI-gated quality thresholds, versioned prompts
- **ML & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn, LSTM & attention architectures, Monte Carlo Dropout, deep ensembles, uncertainty quantification, explainability (XAI)
- **MLOps & Infra:** Git, GitHub Actions (CI/CD), AWS, Linux, Hugging Face Spaces, REST APIs, RAPIDS (GPU-accelerated data science)
- **Databases:** MySQL, PostgreSQL, MongoDB, ChromaDB (vector)
- **Languages & Tooling:** Python, SQL, R, Bash, JavaScript; pandas, NumPy, NetworkX, Matplotlib, Seaborn, Jupyter, ipywidgets, Gradio, LaTeX

PROFESSIONAL EXPERIENCE

Data Analyst Intern — Sudhir Mehrotra & Associates, Chartered Accountants **Jan 2025 – Aug 2025**
Bareilly, India (Hybrid)

- Built Python ETL pipelines automating financial workflows, reducing manual processing effort by 30%.
- Developed a time-series cash-flow forecasting module, improving estimation accuracy by 15% over baseline.
- Automated recurring Excel reporting with Python and VBA, eliminating multi-hour weekly manual tasks for the audit team.
- Designed structured data-reporting systems for audit and compliance teams, improving traceability across reviews.

FEATURED PROJECTS

AskMyDocs — Production-Grade RAG System with CI-Gated Evaluation **Apr 2026**
Python, LLaMA (Groq API), ChromaDB, BM25, Cross-Encoder, GitHub Actions, Gradio, Hugging Face Spaces
Live: huggingface.co/spaces/raghavupadhyay/askmydocs | Code: github.com/raghav-upadhyay2002/AskMyDocs

- Built and publicly deployed an end-to-end RAG app answering questions over user-uploaded PDFs with grounded, citation-tagged responses, using a modular architecture (loader, chunker, embedder, vectorstore, reranker, LLM, prompts) for fast component swaps and testing.
- Engineered hybrid retrieval combining ChromaDB vector similarity (60%) with BM25 keyword scoring (40%), re-ranking top-10 candidates to top-3 with a cross-encoder (ms-marco-MiniLM-L-6-v2).
- Built automated hallucination detection and versioned prompt templates (default / strict / concise) for configurable coverage-vs-grounding trade-offs.
- Built an LLM-as-a-judge evaluation harness scoring faithfulness, relevance, and citation rate, wired into a GitHub Actions CI pipeline that blocks merges below faithfulness 0.70, relevance 0.70, or citation rate 0.80.

LLM-Generated Social Networks — Cross-Cultural Study (M.S. Capstone) **May 2026**
OpenAI API (GPT-4.1), NetworkX, pandas, Matplotlib | Under review at NeurIPS 2026 (arXiv:2605.12898)

- Replicated and extended an ICWSM 2025 study on whether LLMs generate structurally realistic social networks, formalizing four tie-formation mechanisms as conditional distributions over edge sets.
- Built a generation-analysis-benchmark pipeline spanning a 4 prompting-method × 4 culture × 4 language × 3 GPT-4.1-tier × 2-seed matrix (96+ verified conditions, 192 total networks).
- Quantified inter-model divergence via pairwise edge distance (GPT-4.1 ↔ mini = 0.074, GPT-4.1 ↔ nano = 0.119) and identified political affiliation as the dominant homophily dimension under most prompting methods.

Uncertainty-Aware RUL Prediction with LSTM Architectures **Apr 2026**
PyTorch, TensorFlow, scikit-learn, NASA C-MAPSS, ipywidgets
Code: github.com/raghav-upadhyay2002/RUL-prediction-using-LSTM-for-Aircraft-Engine

- Designed and benchmarked four recurrent architectures (VanillaLSTM, StackedBiLSTM, LSTM-Attention, CNN-LSTM) for turbofan remaining-useful-life prediction on NASA C-MAPSS.
- Implemented Monte Carlo Dropout and deep ensembles for uncertainty quantification, using a mean − 2σ lower bound to drive risk-adjusted maintenance decisions instead of point estimates.
- Built gradient-based attribution, temporal attention heatmaps, and an ipywidgets dashboard with tiered maintenance alerts; validated zero-shot cross-dataset transfer across C-MAPSS subsets.

ADDITIONAL PROJECTS

Commonsense Reasoning with Pre-trained Language Models **Dec 2024**

RoBERTa-MNLI, OPT-1.3B, PyTorch, Hugging Face Transformers

Code: github.com/raghav-upadhyay2002/-Evaluation-of-Commonsense-Reasoning-Using-Pre-trained-Language-Models

- Benchmarked RoBERTa-MNLI and OPT-1.3B on commonsense-reasoning datasets, analyzing zero-shot vs. fine-tuned performance across PIQA and related tasks.

Hate Speech Detection **Oct 2024**

Python, scikit-learn, NLP

Code: github.com/raghav-upadhyay2002/Text-Classification

- Built a text-classification pipeline for hate-speech detection using Logistic Regression and Random Forest, reaching 92% accuracy on a Kaggle-sourced dataset to support content-moderation use cases.

CURRENTLY BUILDING

Vision-Based Virtual Maze Navigator **In Progress · Aug 2026**

Python, OpenCV, NumPy, Webots (e-puck robot simulator)

Code: github.com/raghav-upadhyay2002/Vision-Based-Virtual-Maze-Navigator

- Developing an autonomous mobile robot agent that navigates an unknown 3D maze using only first-person camera input in the Webots robotics simulator, targeting a full perceive-map-plan-act autonomy loop (visual SLAM, path planning, closed-loop control).
- Shipped live camera-feed processing and manual teleoperation as scaffolding; currently building an HSV-based visual target detector and wiring it into closed-loop motor control.

CERTIFICATES

- **Fundamentals of Accelerated Data Science — NVIDIA** (Oct 2025): GPU-accelerated computing with the RAPIDS ecosystem, core data science foundations, applied machine learning, accelerated ecosystem integration.
- **AWS Academy Cloud Operations — Amazon Web Services** (Nov 2022): cloud infrastructure operations, monitoring & management, deployment & automation fundamentals.

EDUCATION

University of Arizona **Apr 2024 – May 2026**

M.S. in Data Science · GPA 3.78/4.00 · Tucson, AZ

Relevant coursework: Machine Learning, Applied Natural Language Processing, Computational Linguistics, Data Mining and Discovery

SRM Institute of Science and Technology, India **2020 – 2024**

B.Tech in Computer Science and Engineering (Software Engineering) · CGPA 8.51/10.00

PUBLICATIONS

Kilaru, S.H., Manikyala, S.T., Upadhyay, R., Ramavath, S.S.K., Nunavathu, S., Alharthi, D. “When Do LLMs Generate Realistic Social Networks? A Multi-Dimensional Study of Culture, Language, Scale, and Method.” arXiv:2605.12898, May 2026. Under review at NeurIPS 2026.